

Jeffrey Walker

jfw69@drexel.edu | +1 (445) 214-3307 | github.com/WeffreyJ | linkedin.com/in/jeffrey-walker-4b2314218
| Philadelphia, PA

Summary

Graduate researcher in control systems, autonomy, and robotics with work spanning nonlinear aircraft dynamics, robust control, reinforcement learning, state estimation, SLAM, and embedded experimentation. Strong fit for advanced R&D and research-heavy roles that require rigorous modeling, honest evaluation, and the ability to translate mathematical or simulation work into clear technical evidence.

Research Areas

Control & Learning: nonlinear control, robust control, reinforcement learning, controller comparison, rollout analysis, Monte Carlo robustness analysis

Estimation & Perception: EKF-based sensor fusion, SLAM workflows, trajectory alignment, localization analysis, computer-vision pipelines

Platforms & Tools: MATLAB, Simulink, FlightGear, Python, C/C++, ROS, MuJoCo, OpenCV, Git

Education

Drexel University M.S. Mechanical Engineering, Robotics, Systems & Controls focus 2023–2025
Thesis: *Advanced Flight Control for Nonlinear Aircraft Models (NASA F-18)* Philadelphia, PA

Kwame Nkrumah University of Science and Technology B.S. Automotive Engineering 2019–2023
Project: *Volkswagen Transporter redesign and electrification* Kumasi, Ghana

Research Experience

Graduate Research Assistant – Aircraft GNC & Autonomy *Drexel University, Philadelphia, PA* Jun 2024 – Present

- Modeled and analyzed a 12-state nonlinear NASA-derived F-18 aircraft system across multiple operating regimes for advanced flight-control research.
- Designed and evaluated classical and learning-based controllers including LQR, sliding-mode control, robust-control variants, and reinforcement learning agents.
- Benchmarked controller behavior under disturbance, sensor noise, delays, and actuator saturation, with evaluated runs showing up to **61% reduction in peak state excursions** across primary stability-related states.
- Built validation workflows in MATLAB/Simulink and FlightGear and authored thesis-level documentation covering architecture, assumptions, controller comparisons, and evaluation results.

Aerospace Controller Design Researcher *Drexel University – Prof. Bor Ching Chang, Philadelphia, PA* Jul 2024 – Present

- Supported aerospace controls research involving robust controller design with H_2 and H_∞ methods.
- Contributed to modeling, simulation, and controller-development workflows in MATLAB/Simulink for high-precision aerospace applications.

Industrial Automation Systems Researcher *Drexel University – Prof. Dela Fuentes, Philadelphia, PA* May 2024 – Present

- Conceptualized closed-loop industrial automation architectures using sensing, actuation, and PLC-oriented logic.
- Built prototype hardware using additive and subtractive manufacturing workflows to support controls and automation experimentation.

Selected Research Projects

Nonlinear F-18 Flight Control *Controls / RL / Simulation* 2024 – 2025

- Built a nonlinear aircraft control workflow around a NASA-derived F-18 model using classical control and reinforcement learning methods.
- Organized the project around open-loop versus closed-loop comparison, FlightGear visualization, and technical interpretation instead of abstract AI claims.

AeroPendulum Reinforcement Learning Control *Controls / RL / MATLAB-Simulink* 2025 – 2026

- Built aeropendulum control workflows using reinforcement-learning methods and simulation-backed evaluation tooling.
- Compared learning-based behavior against classical control baselines in a dynamic systems setting relevant to controls and autonomy research.

Sensor Fusion Navigation *Estimation / EKF / Localization* 2025

- Implemented EKF-based fusion of IMU and GPS signals for ground-vehicle navigation under noisy measurements and intermittent dropout.
- Studied localization behavior across representative trajectories with a focus on robustness and estimation quality.

Stereo SLAM Pipeline for UAV Workflows *SLAM / Perception / Evaluation* 2025

- Built a stereo SLAM workflow with KITTI-style data recording, pluggable backend structure, and trajectory evaluation.
- Structured the project around pose visualization, alignment, and interpretation for robotics-navigation experimentation.

Honors & Academic Work

- **Dean's Fellowship**, Drexel University 2023 – 2025
- **Leroy L. Resser Endowment Scholarship** 2024
- **Master's Thesis:** *F-18 Aircraft Dynamical Control Using Classical Control Methods and Reinforcement Learning: A Comprehensive Study*, Drexel University, Aug 2025
- **College of Engineering Student Ambassador**, Drexel University 2024 – 2025